

Geographically Weighted Regression and Causal Inference

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ABSTRACT

Causal inference deals with making reasonable assumptions about causality when direct evidence cannot be collected. An important field of study, many causal inference models exist, each with their own limitations. Geographically Weighted Regression (GWR) is an extension of the basic linear regression model that directly accounts for spatial patterns in geographical data. This paper presents the necessary background for understanding GWR and causal inference modeling, and then provides support for GWR as a causal inference model. This is the first study to make this claim.

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1 INTRODUCTION

A major goal of statistics is to accurately measure and predict the effects of explanatory variables on a dependent variable. Sometimes, causal relationships can be explored through controlled experiments. There are many data sets where this is not possible, like those about the spread of disease, consumer activity, or meteorological phenomena. Causal inference is a procedure whose goal is to better understand causality through alternative means. A causal inference model will need to meet certain key assumptions to produce useful results. [2].

Many of these important data sets contain geographical trends. This means that traditional causal models, like linear regression, which assumes spatial heterogeneity, will fail to capture important local patterns. Geographically Weighted Regression (GWR) is a regression-based approach designed to address this limitation by allowing model coefficients to vary across space and assumes that observations closer in space exert greater influence on one another during estimation [3].

This paper begins by outlining the mechanics of GWR. It then discusses the foundational assumptions required for a model to serve as a valid causal inference framework. Next, it examines the conditions under which GWR can satisfy these assumptions and be interpreted causally. The paper then explores how GWR may be applied to make causal inferences in practice, followed by an evaluation of its strengths and limitations relative to established causal inference methods. To the author's knowledge, this is the first study to systematically investigate GWR as a causal inference framework.

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2 BACKGROUND

2.1 Linear Regression

Geographically Weighted Regression (GWR) extends the classical linear regression model by allowing relationships to vary across space. In the standard linear regression model, the relationship between a response variable y_i and p explanatory variables x_{ik} is assumed to be constant across all observations:

$$y_i = \beta_0 + \sum_{k=1}^p \beta_k x_{ik} + \epsilon_i, \quad (1)$$

where $\beta_0, \dots, \beta_p$ are global parameters and ϵ_i is a random error term with zero mean. Model parameters are estimated using ordinary least squares,

$$\hat{\beta} = (X^T X)^{-1} X^T Y. \quad (2)$$

While this model is useful in many cases, the assumption of spatially invariant coefficients limits its suitability for data with spatial heterogeneity [3].

2.2 Geographically Weighted Regression

Geographically Weighted Regression (GWR) generalizes linear regression by allowing regression coefficients to vary spatially. Let y_i denote the response variable, x_{ik} the k th explanatory variable, and (u_i, v_i) the spatial coordinates of observation i . The GWR model is

$$y_i = \beta_0(u_i, v_i) + \sum_{k=1}^p \beta_k(u_i, v_i) x_{ik} + \epsilon_i, \quad (3)$$

where $\beta_k(u_i, v_i)$ are location-specific coefficients and ϵ_i is a mean-zero error term.

Local parameter estimates are obtained via weighted least squares. Let X denote the $n \times (p + 1)$ design matrix and Y the response vector. The estimator at location (u_i, v_i) is

$$\hat{\beta}(u_i, v_i) = \left(X^T W(u_i, v_i) X \right)^{-1} X^T W(u_i, v_i) Y, \quad (4)$$

where $W(u_i, v_i)$ is a diagonal matrix of spatial weights.

3 CAUSAL INFERENCE MODELING

A causal inference model needs more than a just a predictive formula like GWR. It needs a causal framework that connects data to interventions and counterfactuals. A causal inference model needs the following things

- (1) A structural causal model
- (2) Additional assumptions
- (3) Identification strategy

Structural Causal Models

A structural causal model is a mathematical model that represents a causal system. It contains a set of variables, a directed causal graph, a structural equation for each variable, independent noise terms, and intervention semantics.

- (1) Variables: observed and unobserved factors in the system
- (2) Directed causal graph: a graph that says which factors cause which other factors

- (3) Structural equations: an equation for each variable where inputs are all its causing factors, as well as the noise factor. So, if we have causal graph $X_1 \rightarrow X_2$, then we have equation $X_2 = f(X_1, N_2)$
- (4) Independent noise term: each variable has its own independent noise term in its structural equation
- (5) Intervention semantics: we can alter structural equations to support interventions without changing other mechanisms. So if for the previous setup, if $X_1 = 5$, then $X_2 = f(5, N_2)$.

[11]

Causal Assumptions

For causal inference methods to yield meaningful results, a set of assumptions about the data and causal structure of the model must be satisfied. When these assumptions are violated, causal estimates may become biased, unstable, or uninterpretable. In practice, these conditions are rarely exactly met; however, if these assumptions are approximately met or are reasonably justified, it is still possible to determine useful results from causal analysis. These key assumptions vary, but typically include positivity, identifiability, exchangeability, modularity, consistency, and SUTVA/SUNTVA [9].

- (1) **Positivity:** A causal model satisfies the *positivity assumption* if there are strictly positive probabilities for treatment assignment across all covariate values. When positivity holds, treatment and control groups overlap across the range of covariates. This allows causal effects to be identified with observed observations. When there is not sufficient overlap, estimation of causal effects requires extrapolating into unknown regions of data, which can lead to unreliable and unstable estimates.
- (2) **Identifiability:** A causal model satisfies *identifiability* if the causal effect can be computed unambiguously from observed data, given the causal structure. This means that the observed data and the causal structure determine a unique value, no alternative values can be found with the same information. For identifiability to hold, two conditions must be met.
 - (a) A sufficient amount of appropriate data must be available. An insufficient sample size may result in unstable or biased estimates. In addition, the data must satisfy the positivity assumption.
 - (b) The causal model must control non-causal paths between the treatment and the outcome. Non-causal pathways are when a variable is statistically associated with the outcome, but does not directly cause the outcome. This is commonly caused by confounding, which is when a variable influences both the treatment and the outcome. If left unaddressed, these can cause bias and error.
- (3) **Exchangeability:** A causal model satisfies *exchangeability* if its treated subjects, had they not received the treatment, would have experienced the same average outcome as the untreated subjects who did not receive the treatment, and vice versa. That is, treatment assignment should not systematically depend on factors that also affect the outcome. Exchangeability is assumed to be achievable through unconfoundedness. It holds when all relevant confounders are observed and appropriately controlled for using deconfounding techniques.
- (4) **Modularity:** A causal model satisfies *modularity* if all causal mechanisms operate independently. Interventions affect only the causal mechanisms directly targeted by the intervention, while all other mechanisms in the system remain unchanged. Without modularity, interventions could cause unforeseeable changes throughout the entire system, making causal reasoning difficult.
- (5) **Consistency:** A causal model satisfies *consistency* when each level of all treatments correspond to a single, well-defined intervention. Violations of consistency occur when substantially different versions of a treatment

are grouped together and treated as equal. Each would produce different outcomes if they had been given to the same subject. [9]

- (6) **SUTVA**: A causal model satisfies *the Stable Unit Treatment Value Assumption* (SUTVA) if these two conditions are met
- (a) The model meets the consistency assumption
 - (b) No interference between units: the treatment assigned to one unit does not affect the outcome of any other unit. The potential outcome for a given unit depends only on that unit's own treatment and not on the treatment's of other units. This assumption rules out cross-unit influence which ensures that potential outcomes are well defined and can be attributed unambiguously to treatment assignments.

In many real-world applications, the no-interference component of SUTVA is often violated. To address this, extensions of SUTVA have been proposed, like *the Stable Unit Neighborhood Treatment Value Assumption* (**SUNTVA**), which allows outcomes to depend on treatments within a predefined neighborhood. It is still assumed that interference does not extend beyond this neighborhood.

[2]

Identification Strategy

If a predictive model was asked to estimate the effect of a change of one of its predictor variables, its response may not reflect actual causality. It may just reflect correlation or association. I.e. if it is predicting the price of a house given an increase in number of bedrooms, its prediction may be correlated with houses with higher square footage (which also have a higher number of bathrooms), rather than the same house with an additional bathroom. An identification strategy is needed to explain why a model finds causality, rather than correlation.

Identification strategies work by creating or finding variation in a variable that is not mixed up with confounders, so outcome differences can be interpreted as causal. Typically, this is done in one of three ways

- (1) Make treatment random: By having randomized controlled trials, we can compare before and after outcomes to see the average causal effect. However, having controlled trials is often not possible.
- (2) Holding confounder fixed: We only compare units that have the same confounders, but differ with treatment. This can be done by matching. In this case, we are controlling for confounders directly.
- (3) Assignment structure: Use a rule exploiting the data's structure to make nearby units comparable. This is done when treatment isn't random, but around a boundary its effectively random.

4 GWR AS A CAUSAL INFERENCE MODEL

For GWR to serve as, or in, a model for causal inference with a given data set, it has to have the three criteria listed in section 3. That is

- (1) A structural causal model
- (2) Additional assumptions
- (3) Identification strategy

Firstly, it is important to note that many causal definitions are stated in terms of discrete and finite variables. While, this formulation work well for categorical data, many practical applications involve continuous variables. In such cases, the assumptions must be interpreted in an approximate sense. Further, assume that all data is spatial, and that geographic location is an influential variable for all covariates, but not for the response variable itself.

Data

For any given data set, a structural causal model can be drawn. This includes structural equations and mechanisms. It cannot be guaranteed that equations are right, directions are right, or that all confounders have been identified. This will have to be assumed. For many data sets, this is intuitive. For others, multiple models may have to be built and tested to determine such. Note that every structural equation will include geographic location as a variable.

The positivity, consistency, and modularity assumptions can be checked with just the data sets. To identify positivity in a spatial dataset, each covariate should be mapped. Then, they should be checked for overlapping regions. This gets exponentially harder with more variables. Consistency can be reasoned for each variable. Each variable pair should be reasoned for modularity both ways.

GWR Captures Spatial Variation

Recall equations 3 and 4. Left undefined earlier was W , the weight matrix, which is determined by a kernel function of distance. A common choice is the Gaussian kernel,

$$w_{ij} = \exp\left(-\frac{d_{ij}^2}{2b_i^2}\right), \quad (5)$$

where d_{ij} is the distance between observation j and location (u_i, v_i) .

b is an adaptive bandwidth, which is governed by a global nearest-neighbor parameter k . For each location (u_i, v_i) , the bandwidth b_i is defined as the distance from (u_i, v_i) to its k th nearest neighbor. As a result, the spatial scale of the model adapts to data density: in densely sampled regions b_i is small, while in sparsely sampled regions b_i becomes larger. The Gaussian kernel then uses b_i to control the rate at which weights decay with distance.

The optimal bandwidth parameter k is selected by minimizing the corrected Akaike Information Criterion (AICc), given by

$$\text{AICc} = 2n \log(\hat{\sigma}) + n \log(2\pi) + n \left(\frac{n + v_1}{n - 2 - v_1} \right), \quad (6)$$

where $v_1 = \text{tr}(S)$ is the effective number of parameters and S is the GWR hat matrix. The i th row of S is given by

$$S_i = X_i \left(X^T W(u_i, v_i) X \right)^{-1} X^T W(u_i, v_i), \quad (7)$$

where X_i denotes the i th row of the design matrix X , and $\hat{\sigma}^2 = \text{RSS}/(n - v_1)$. The AICc provides a trade-off between model fit and complexity, with smaller values indicating a better model.

Spatial heteroskedasticity may be addressed by extending the standard GWR framework to incorporate local variance-based reweighting. Residuals are first obtained from an initial GWR fit, and a local variance estimate σ_i^2 is computed using the k nearest residuals around location (u_i, v_i) . An additional weight is then defined as

$$aW(u_i, v_i) = \frac{1}{\sqrt{\sigma_i^2 + \theta}}, \quad (8)$$

where $\theta > 0$ prevents division by zero. These weights are multiplied with the spatial weights and the model is iteratively refit until convergence, reducing the influence of high-variance regions.

Outliers may also distort local estimates. In a spatial context, outliers are defined relative to local neighborhoods rather than the full dataset. Externally studentized residuals are therefore used, defined as

$$r_i^* = \frac{e_i}{\hat{\sigma}_{-i} \sqrt{q_{ii}}}, \quad (9)$$

where e_i is the residual for observation i , $\hat{\sigma}_{-i}$ is the standard deviation estimated with observation i removed, and q_{ii} is the (i, i) th element of

$$Q = (I - S)(I - S)^T, \quad (10)$$

with S denoting the GWR hat matrix. The variance estimate is given by

$$\sigma^2 = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n - 2v_1 + v_2}, \quad (11)$$

where $v_1 = \text{tr}(S)$ and $v_2 = \text{tr}(SS^T)$. Observations with large studentized residuals may be downweighted or removed [5].

Spatial Confounding and GWR

Identifiability and exchangeability both relate to controlling confounding. GWR can do this, but only under the assumption that all confounders vary smoothly in space. GWR directly accounts for spatial variation of covariates, which would include variation caused by confounders. These confounders must be smooth, as GWR's bandwidths are continuous, so discrete jumps could not be picked up on.

Spatial data often assumes that there is interference between units. GWR is uniquely designed to confront this. Note from equation 5 that observations outside of an optimized bandwidth from a given observation have very little weight on that observation's estimated response. This means that outcomes depend on treatments within a predefined neighborhood, which means that GWR meets the SUNTVA assumption.

GWR works as an identification strategy. Note that only observations within a neighborhood of a given observation will have significant effect on that observation's predicted response. Since confounders vary spatially, the effects of confounders within this neighborhood can be considered close enough to being identical. This effectively allows for condition on confounders to get clear, causal results from GWR instead of associative for each prediction.

With the right data set, GWR can effectively work as a model for causal inference. The data set must have a reasonable structural causal model; meet the positivity, consistency, and modularity assumptions; and only contain confounders that smoothly vary in space. With these conditions, GWR meets all criteria to be considered a valid and useful model for causal inference. With equation 3, GWR can easily predict a counterfactual. Performing this on an entire dataset can find the effects of an intervention. This operation can find the average treatment effect (ATE) and inverse probability weighted ATE (IPW ATE). These are primary objectives of causal inference.

5 IMPLEMENTATION

The goal of this project was to create a program to run GWR on a dataset, evaluate how well it runs, and then let the user find causal results. It is implied that the user has assessed the dataset before running the program to verify that the data meets all assumptions it should to be used in a GWR causal inference model. This program was coded in python on a jupyter notebook. It makes use of a number of pre-existing libraries, like numpy, pandas, scipy, sklearn, and several others.

The user begins the program by passing in an excel file with all the data, structured so that the first column is the dependent output, the next two are latitude and longitude, respectively, and the remaining columns are explanatory variables. The user also specifies the number of explanatory variables, as well as the number of observations from the data set that they would like to regress on, and if the output data is continuous (opposed to Poisson or binomial data).

After building several numpy arrays of data, the program performs kernel density estimation (KDE) on the output variable. This maps the shape of the output data, allowing us to both verify that it is continuous, and to see if it is right or left skewed. If the data is skewed, it is then appropriately log transformed. This can significantly help with prediction accuracy. It is implied that the user will recall this when viewing prediction results, as some will be returned exponentiation.

The program uses a test-train split to first train the GWR model, as was described in earlier sections, before testing and evaluating the model on several criteria that will be discussed in the next section. This allows users to decide if the model has appropriately captured causal insights or not. If so, then can proceed.

From here, the user can find three core causal insights: counterfactuals, interventions, and average treatment effects. Counterfactuals are found by giving the model a hypothetical observation to estimate the output value of. Interventions are found by using the counterfactual function on every observation with the same value for a given variable/s. This is also mapped, and the before and after averages of their outputs are found. Before finding the ATE of a given variable, the program checks if the given variable is binary. If not, it is assigned a binary value reflecting if a given observation's value for that variable is above or below the global average. Propensity scores are then found to calculate the inverse probability ATE.

6 EVALUATION

GWR is a predictive model, so it could be compared to other predictive models, like linear regression or neural networks, through metrics such as mean-squared error for accuracy of predictions, Moran's I for spatial autocorrelation of residuals, or R^2 /coefficient of determination for proportion of variance explained by the model. However, the purpose of this paper is not to evaluate this GWR model for its predictive capabilities. Instead, we wish to see how it performs as a causal inference model.

Most causal inference models do not directly makes predictions. Instead, they aim to estimate the effects of treatments, which can be called a counterfactual. For this reason, different metrics are needed to compare GWR to other causal inference models, like causal forests or propensity matching. One could still do predictive comparisons to confirm that GWR effectively capture the spatial patterns of a given dataset. If GWR performs similar to a linear regression model, then the data is likely not spatial. If GWR is significantly outperformed by a more complex model, like a neural network, then it likely does not sufficiently capture spatial trends in the data.

Many of the most important ways to evaluate a causal inference model involve assessing the quality of its estimations of treatments. The only way to know the quality of an estimator is to compare its estimations to their actual values. This is impossible for real datasets. The fundamental problem of causal inference is that we can only observe an outcome under the single treatment that actually occurred; there is no way to observe the outcome under a treatment that did not occur.

This can be addressed by generating data sets. It is possible to take an existing dataset, simulate a treatment assignment mechanism including confounding, then generate outcomes for treated and untreated observations. This is called semi-synthetic data. If the original data is spatial and the treatment assignment mechanism and confounding are both spatial, then the final data set is still spatial. It would be possible to do GWR on such a data set, allowing for comparison.

6.1 Data Generating

The general approach to synthetic data generating is given in Heiss [7]. The specific approach given here was formulated by the author.

Take a spatial data set which is considered valid for GWR use. Covariates, outputs, and coordinates are standardized for calculations. Covariates are kept standardized. A new variable called confounder is made. For each observation, we have

$$U_i = 0.45y_i + 0.20\sin(1.5lat_i)\cos(1.2lon_i) + \eta_i \quad (12)$$

where $\eta_i \sim N(0, 0.6)$. U is then standardized. U is intuitively spatial. Then, we find the propensity score, p_i , for a given observation where

$$\log\left(\frac{p_i}{1-p_i}\right) = \alpha + 0.90y_i + 0.35x_{i1} + 0.25x_{i2} - 0.20x_{i3} + 0.25(x_{i1}x_{i2}) + 0.35\sin(1.1lat_i) - 0.25\cos(1.0lon_i) + 0.65U_i \quad (13)$$

where α is the treatment intercept and logits were clipped to keep values more reasonable. With this probability, treatment is binomially assigned $T_i \sim \text{Bernoulli}(p_i)$. This makes treatment spatial. Finally, the baseline outcome function is

$$f_i = 1.00y_i + 0.30x_{i1} + 0.20(x_{i2}^2) + 0.15x_{i3} + 0.25\sin(1.2lat_i)\cos(0.8lon_i) + 0.35U_i \quad (14)$$

where $\eta_i \sim N(0, 0.6)$, $\tau = 0.8$, $Y_{i0} = f_i + \eta_i$, and $Y_{i1} = f_i + \tau + \eta_i$.

Five such data sets have been chosen and transformed. They each meet assumptions previously defined. The data sets are called Crime, Depression, HIV, Housing, and Mental_health. Each contains information about rates of their respective title, plus explanatory covariates across a region of the United States. Housing is about housing prices and their covariates. Each has at least a few thousand observations and 6-13 explanatory variables.

6.2 Causal Metrics

Average treatment effect (ATE) error evaluates how accurately a model estimates the average treatment effect across the population. Let the true individual treatment effect be $\tau_i = Y_i(1) - Y_i(0)$ and the model's predicted effect be $\hat{\tau}_i$. The ATE error is

$$ATE_{\text{error}} = \left| \widehat{ATE} - ATE \right| = \left| \frac{1}{n} \sum_{i=1}^n \hat{\tau}_i - \frac{1}{n} \sum_{i=1}^n \tau_i \right| \quad (15)$$

which is small when the model recovers the correct population-level effect [10].

Precision in estimation of heterogeneous effect (PEHE) measures how well the model recovers individual-level treatment effect heterogeneity. Using the same notation, PEHE is the root mean squared error between predicted and true individual treatment effects:

$$PEHE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{\tau}_i - \tau_i)^2}. \quad (16)$$

A lower PEHE indicates the model is accurately capturing which observations have larger or smaller treatment effects. Importantly, a model can have low ATE_{error} but high PEHE if it gets the overall average effect right while misestimating treatment effects for specific individuals [4].

6.3 Causal Comparisons: GWR vs Propensity vs Forest

Below are the results from running the 5 data sets with GWR, and two standard causal inference models: Causal Forest (CF), and Propensity Scoring (PS). All models were run with only 1000 observations.

Across all datasets, GWR consistently achieves the lowest or near-lowest ATE error and PEHE compared to causal forests (CF) and propensity score (PS) methods, indicating more accurate estimation of both average and individual treatment effects. This performance advantage is especially pronounced in datasets like Depression and HIV, where

Dataset	model	Time	ATE error	PEHE
Crime	GWR	54s	0.359	0.372
Crime	CF	6s	0.367	1.05
Crime	PS	2s	0.399	4.47
Depression	GWR	51s	0.043	0.255
Depression	CF	9s	0.612	0.802
Depression	PS	2s	1.14	5.26
HIV	GWR	52s	0.047	0.261
HIV	CF	8s	0.572	1.27
HIV	PS	2s	2.21	24.4
Housing	GWR	68s	0.230	0.305
Housing	CF	10s	0.244	0.575
Housing	PS	5s	0.535	3.749
Mental Health	GWR	106s	0.289	0.364
Mental Health	CF	22s	0.320	0.616
Mental Health	PS	18s	0.778	142.35

Table 1

GWR’s errors are substantially smaller than those of the competing models. CF generally performs second-best, often with moderate ATE error but noticeably higher PEHE, suggesting it captures average effects reasonably well but struggles with individual-level heterogeneity. PS methods perform worst overall, with significantly larger errors—particularly in PEHE—indicating poor recovery of individual treatment effects. However, this improved accuracy from GWR comes at a cost: its runtime is consistently much higher than the alternatives. GWR’s time signature is $O(n^2)$, which is significantly worse than the other models, which are between $O(n \log n)$ and $O(n)$. Overall, the results highlight a clear trade-off, where GWR offers superior causal accuracy at the expense of time and resources.

7 RELATED WORKS

7.1 Estimating Direct and Indirect Causal Effects

Ali et al [2]. propose a deep learning–based causal inference framework designed to estimate both direct and indirect effects in the presence of spatial interference. Their method modifies the potential outcome framework to spatiotemporal settings by modeling how treatments at one location influence neighboring locations, violating the standard SUTVA assumption. They adopt the SUNTVA, allowing outcomes to depend on those in surrounding regions. Their model uses a latent factor to handle time-varying confounding with a network that captures local and global spatial dependencies. This enables the estimation of direct treatment effects, indirect spillover effects, and lagged effects across time.

7.2 Causal Inference for Spatial Treatments

Pollmann [12] proposes a causal inference framework designed for spatial treatments, where treatments occur at locations and affect nearby units. Instead of comparing treated units to nearby untreated units, his method compares individuals near observed treatment locations to individuals near counterfactual (unrealized) treatment locations, addressing biases in common spatial comparisons. He makes inverse probability weighting estimators and standard errors that remain valid even under spatial correlation, avoiding the need to model spatial dependence. He assumes spatial unconfoundedness and uses convolutional neural networks to identify comparable counterfactual locations based on spatial covariate patterns.

7.3 Spatial Causality

Akbari et al. [1] review spatial causal inference, outlining the key challenges and methods used in the field. They emphasize that traditional causal inference frameworks, like the potential outcome model, often fails in spatial settings due to spatial dependence and violations of SUTVA. The paper categorizes common causal approaches used in spatial data, including matching methods, difference-in-differences, and structural causal models. Their approach is not a single model, but rather a framework summarizing the current state of the field. They identify theoretical limitations and challenges that any spatial causal model must address.

8 CONCLUSIONS

Spatial data underlies complex real-world phenomena, yet introduces challenges that traditional causal methods often fail to solve. This paper examined Geographically Weighted Regression (GWR) as a framework for causal inference. We demonstrated that, under appropriate assumptions, it can be used to estimate treatment effects and generate meaningful counterfactual insights in spatial settings.

When a valid structural causal model is specified, key assumptions such as positivity, exchangeability, and consistency are reasonably satisfied, and confounders vary smoothly across space, GWR can function as both a modeling and identification strategy. Its local weighting mechanism allows for comparisons between nearby observations, controlling for spatially structured confounding and allowing for estimation of localized causal effects. GWR can then be used in practice to estimate counterfactuals, interventions, and average treatment effects. Results on semi-synthetic spatial datasets show that GWR performs comparably to established causal inference methods such as causal forests and propensity score approaches. Additionally, GWR offers a significant advantage in interpretability, as it provides spatially varying coefficients that can be directly analyzed, unlike many black-box models.

However, limitations remain. GWR is computationally intensive. It has a quadratic time complexity that limits scalability to large datasets. Its validity also depends on confounders varying smoothly in space, which is strong assumption to make. Assessing whether a dataset satisfies causal assumptions remains a challenging and largely manual process. Future work should focus on improving computational efficiency and developing automated methods for validating causal assumptions. Hybrid approaches that combine the interpretability of GWR with the flexibility of modern causal machine learning methods may provide a promising direction for advancing spatial causal inference.

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